

**ASSIGNMENT 4: Clustering and Regression****Prof. Varun Dutt****Due: Before 11:59 PM on Sep 8, 2026**

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**Readings:**

- Class notes and slides from week 7 and 8
- Witten, Frank, and Hall (2011):  
[https://www.dropbox.com/s/j5jxyfiw1nsv5xy/%5Blan H. Witten%2C Eibe Frank %2C Mark A. Hall%5D Data Min%28BookFi.org%29.pdf?dl=0](https://www.dropbox.com/s/j5jxyfiw1nsv5xy/%5Blan%20H.%20Witten%20Eibe%20Frank%20Mark%20A.%20Hall%5D%20Data%20Mining%28BookFi.org%29.pdf?dl=0)
- Weka website: <http://www.cs.waikato.ac.nz/ml/weka/>

**Objectives:**

- Clustering in datasets using Hierarchical and k-means algorithms.
- Linear and Logistic Regression in WEKA.

**Assignment:**

1. Please see recent Hollywood movies with the following attributes: [12]

| Patterns                                 | Attributes                  |                               |
|------------------------------------------|-----------------------------|-------------------------------|
|                                          | Total Gross<br>(Million \$) | Max Theater Count<br>(Number) |
| Mission:<br>Impossible -<br>Rogue Nation | 49                          | 3988                          |
| The Intern                               | 18                          | 3305                          |
| A Walk in the<br>Woods                   | 27                          | 2158                          |
| Jurassic World                           | 650                         | 4291                          |
| Minions                                  | 334                         | 4311                          |

Note. The source of this information is the following:

<http://www.boxofficemojo.com/>

- A. Please apply the hierarchical clustering procedure to the dataset above and derive 2 appropriate clusters. For this purpose, please use the single-linkage (or nearest neighbor) procedure.

- B. Now, apply the same hierarchical clustering procedure; however, please use the centroid-linkage procedure.
- C. Furthermore, now apply the k-means algorithm with  $k = 2$  and perform clustering. For the purpose of applying the clustering, please use the Mission Impossible and Minions as the starting clusters.
- D. Now, watch this video on evaluating clustering:  
<https://www.youtube.com/watch?v=9aODdNSAaul> (please pay particular attention to the classes to cluster evaluation. Open Weka and the weather.numeric.arff dataset and perform k-means clustering on the classes to cluster evaluation. Please report the accuracy of the clusters mapped to classes. Now, change the seed = 10 to seed = 11 in the settings window of k-means clustering. Does the accuracy change upon running the classes to cluster procedure again? If yes, then why yes; and, if no, then why no?

2. Now, consider the Hollywood movies dataset given below:

[6]

| Patterns                                    | Attributes                  |                                  |                        |                                           |
|---------------------------------------------|-----------------------------|----------------------------------|------------------------|-------------------------------------------|
|                                             | Total Gross<br>(Million \$) | Max Theater<br>Count<br>(Number) | Budget<br>(Million \$) | Number of Weeks<br>running in<br>Theaters |
| Mission:<br>Impossible -<br>Rogue<br>Nation | 49                          | 3988                             | 150                    | 9                                         |
| The Intern                                  | 18                          | 3305                             | 35                     | 1                                         |
| A Walk in<br>the Woods                      | 27                          | 2158                             | 22                     | 4                                         |
| Jurassic<br>World                           | 650                         | 4291                             | 150                    | 16                                        |
| Minions                                     | 334                         | 4311                             | 74                     | 12                                        |

- A. Please use Weka and import the above dataset in an .arff file in Weka. Considering the Total Gross as the dependent variable and other variables as independent variables (predictors), develop a linear regression model on the above dataset. What is the regression model developed? Report the accuracy of the linear regression model (i.e., the error terms) and the F-measure?
- B. Now, consider the following test sample:

| Patterns    | Attributes                  |                                  |                        |                                           |
|-------------|-----------------------------|----------------------------------|------------------------|-------------------------------------------|
|             | Total Gross<br>(Million \$) | Max Theater<br>Count<br>(Number) | Budget<br>(Million \$) | Number of Weeks<br>running in<br>Theaters |
| The Martian | ?                           | 3831                             | 108                    | 1                                         |

Using the regression model developed, please predict the total gross in m\$ earned by movie above. Now, open the website <http://www.boxofficemojo.com/weekend/> and go to the October 2<sup>nd</sup> – 4<sup>th</sup> weekend (the link will be posted on Monday, 5<sup>th</sup> October, 2015). Please use the actual Total Gross (in m\$) given on the website for The Martian and indicate the relative error:  $ABS (Actual\ Total\ Gross - Predicted\ Total\ Gross) / Actual\ Total\ Gross$ , where Predicted Total Gross is from the Regression model and Actual Total Gross is from the boxoffice website. What is the percentage in your predicted estimate?

3. Consider the weather.nominal.arff dataset in Weka. Open the dataset and choose the logistic regression algorithm (under functions) to this dataset. Use 10-fold cross validation while running the logistic regression model. Also, under more options, please output the predictions. Now, run the model and provide the proportion of correctly classified instances (also provide the F-measure). Please report the training patterns where the logistic regression makes an error in classification. [4]

Please ensure that the names of all team members are on your submission! Please submit a word/PDF document with your responses to the above questions. In addition, please submit your .ARFF files for the training and test sets in question 2. Please submit your assignment to [LMS](#)